# Cellular Energetics: Photosystem II and ATP Synthesis

## 1. Light-Dependent Reactions in the Thylakoid Membrane

Photosynthesis converts solar photons into chemical potential energy stored in ATP and NADPH. Within chloroplast grana, chlorophyll pigments absorb light at wavelengths 680 nm (Photosystem II) and 700 nm (Photosystem I).

### Figure 4.1: Water Photolysis and Electron Transport Chain

Enzymatic splitting of water molecules releases electrons and protons into the thylakoid lumen:

### $$2H_2O \rightarrow 4H^+ + 4e^- + O_2$$

The resulting proton electrochemical gradient drives ATP Synthase rotor kinetics, generating ATP from ADP and inorganic phosphate.